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Stakeholders:

Designation	Name	Signature
Faculty Members		
Student Researcher (Final-Year Undergraduate)	Redwan Hassan (2021831047)	
University Administration		
Research Office		
External Researchers		

Business Requirements Specification

University Research Collaboration Center (URCC)

1. Introduction

The University Research Collaboration Center (URCC) is a system which will improve research collaboration among internal and external researchers by automating the entire process. It has central research information, collaboration requests, and shared activities inside one platform.

1.2 Purpose

This document identifies and describes the business requirements of the URCC. It helps to understand the stakeholder needs, system scope and guides the future development of the system without focusing on the technical side.

2. Problem Statement (Multiple Stakeholder Stories)

2.1 Student Perspective

Students interested in research often struggle to find suitable projects and supervisors. They rely on informal communication such as emails or personal contacts, which sometimes cause late responses or missed opportunities.

2.2 Faculty Perspective

Faculty members receive many unstructured research requests through emails. Due to the lack of a centralized system, it becomes difficult to evaluate students, manage supervision capacity, and track ongoing research.

2.3 External Researcher Perspective

External researchers and industry collaborators face difficulty choosing appropriate university contacts. The absence of a formal platform reduces collaboration opportunities and slows partnership development.

2.4 Administrative Perspective

University administration lacks a unified view of research activities across departments. This makes it challenging to monitor collaboration levels, measure research productivity, and support institutional research goals.

These problems visited from different stakeholders perspectives require solution with a centralised platform.

3. Project Objective

- i) Provide a centralized platform for research collaboration
- ii) Improve communication among students, faculty, and external researchers
- iii) Increase transparency of research opportunities
- iv) Enhance overall research productivity and coordination

4. Scope

4.1 In Scope

- i) Publishing research interests and project opportunities
- ii) Allowing students to search and apply for research projects
- iii) Managing collaboration requests
- iv) Supporting collaboration with external researchers
- v) Providing research activity visibility to university administration

4.2 Out of Scope

- i) Conducting actual research work
- ii) Managing research funding and grants
- iii) Handling publication submissions
- iv) Performance evaluation of researchers
- v) Detailed project task management

5. Stakeholders

- i) University Administration
- ii) Research Office
- iii) Faculty Members
- iv) Students
- v) External Researchers

6. User Stories

- **Finding Collaborators**

As a student researcher, I want to find collaborators inside and outside the university through a single platform so that I do not rely on informal methods.

- **Matching interests between collaborators**

As a student, I want the system to match me with collaborators based on research interests and skills so that I can join relevant projects efficiently.

- **Use of Statistics and Analytics**

As a student, I want analytics such as response rates and skill gaps so that I can improve my chances of research acceptance.

- **Clean User Interface**

As a student, I want publications, projects, and funding opportunities to be displayed in a structured and simple way so that I can quickly understand and apply.

- **User Engagement**

As a faculty member, I want to publish research opportunities on a centralized platform so that interested students can apply in a structured manner.

- **Request Validation and Review**

As a faculty member, I want to review and manage student collaboration requests so that I can control my supervision workload effectively.

- **Collaborator Discovery**

As an external researcher, I want to identify faculty members and research groups within the university so that collaboration opportunities can be established easily.

- **Automatic Review and Summary Generation**

As a university administrator, I want to view summaries and reports of research collaborations so that I can monitor research activities and institutional performance.

Appendix A: Image of Documents

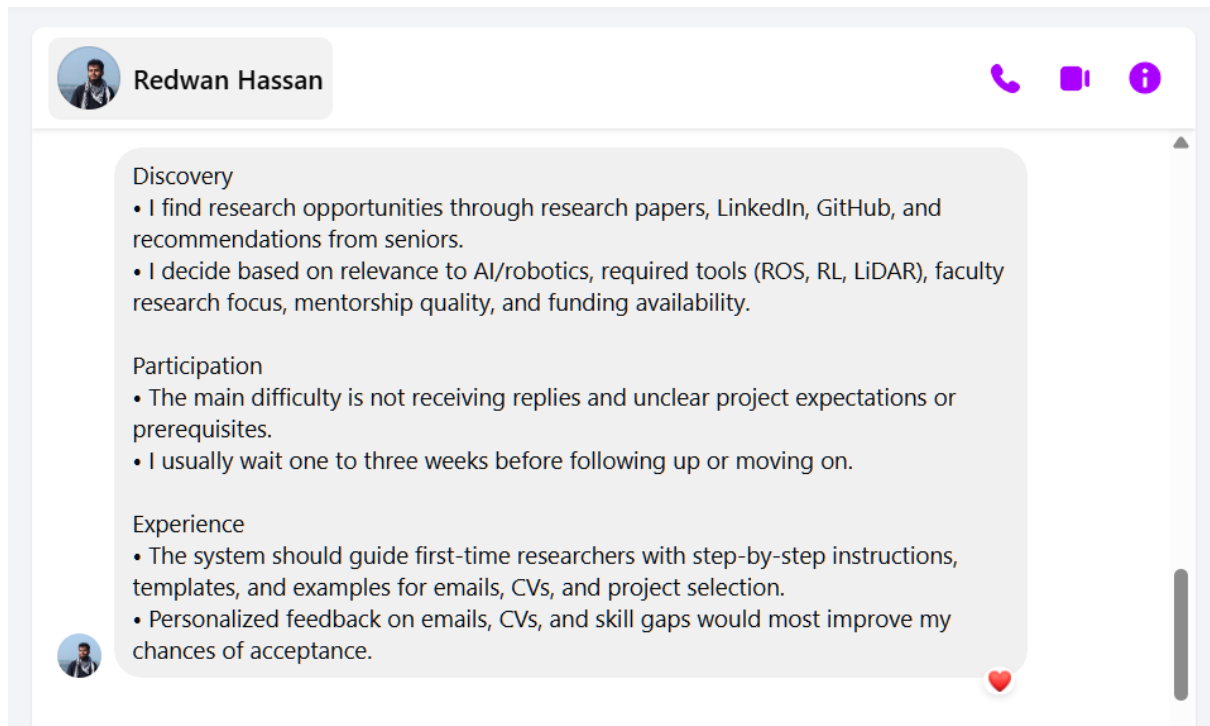


Figure A.1: Interview of Student Stakeholder

The image represents the interview conducted by our team with the student stakeholder. He shares his opinions, insights on the project.



Figure A.2: Research center of SUST

The photo shows the research center of Shahjalal University of Science and Technology. Our team visited the center with a view to take an interview of researchers regarding the project.

Appendix B: Student Stakeholder Interview Summary

Stakeholder Type: Student Researcher

Interview Method: Semi-structured interview

Discovery

- Students currently discover research opportunities through research papers, professional platforms such as LinkedIn and GitHub, and recommendations from senior students.
- Decision-making is influenced by research relevance (e.g., AI and robotics), required tools and technologies, faculty research focus, mentorship quality, and availability of funding.

Participation

- The primary challenges faced by students include lack of response from faculty members and unclear project expectations or prerequisites.
- Students typically wait one to three weeks before following up or abandoning a research opportunity.

Experience

- Students expressed the need for structured guidance for first-time researchers, including step-by-step instructions, templates, and examples for emails, CVs, and project selection.
- Personalized feedback on communication, CV quality, and skill gaps was identified as a key factor that could improve acceptance chances.